

Worksheet — Hanging on Two Threads Simulation

Use with *Simulation - Hanging on two threads.html* · one mass on two threads tied to two fixed supports · equilibrium of three forces (Blocks 3–4)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open physics.taneski.com/24-hanging-two-threads/ in a browser (offline: *Simulation - Hanging on two threads.html*). Choose an example at the top: 🧪 Lab · 🐦 Bird · 🚶 Walker · 🏮 Lantern · 🚦 Traffic · 📰 Sign · 🧺 Washing. The **Example** list has more set-ups for each one. Set m , θ_L , θ_R and the thread with ▼▲ or type a value. Press ▶ **Let go**: the mass hangs at rest, or a thread snaps.



Scan to open

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 25 min) · **Part 3** homework (≈ 30 min).
 $g = 9.81 \text{ N kg}^{-1}$. Answers to 3 s.f.

Angles θ_L and θ_R are measured **below the horizontal**. At rest: $T_L \cos \theta_L = T_R \cos \theta_R$ and $T_L \sin \theta_L + T_R \sin \theta_R = mg$.

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🧪 A 2.0 kg mass hangs from the middle of two threads, each at 30° below the horizontal. Is the tension in each thread half the weight, equal to the weight or more than the weight?			
2. 🧪 You make both threads flatter (smaller θ). Does the tension increase, decrease or stay the same?			
3. 🧪 Can the threads be pulled perfectly straight ($\theta = 0^\circ$) with the mass still hanging on them?			
4. 🐦 A 30 g bird lands on a cotton thread (breaks above 20 N) pulled almost straight: 0.4° each side. Does the thread hold?			
5. 🚦 A traffic light hangs from cables at 30° (left) and 60° (right). Which cable has the larger tension?			

6.  A tightrope walker stands near the left end: 12° on the left, 3° on the right. Which part of the wire has the larger tension — by a lot or a little?			
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7. Calculate the tension in each thread when a 2.0 kg mass hangs at 10° on both sides. Check:  Lab → Example 2.0 kg , both threads at 10° .

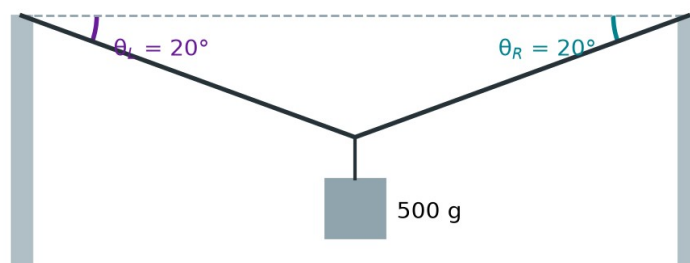
Working: FBD of the mass → $\Sigma F_y = 0$

Part 2 — During the lab: a string between two stands

Apparatus: two clamp stands with bosses (same height), strong string, two newton meters ($0\text{--}10\text{ N}$), 500 g mass hanger, protractor, metre rule.

Method: tie the string between the stands with a newton meter in each half; hang the 500 g mass from the middle. Move the stands apart or together to change the angle. Measure each angle below the horizontal with the protractor and read both newton meters.

 **Safety:** keep feet away from under the mass; clamp the stands to the bench (or hold the bases) — they are pulled towards each other. Do not go below about 10° : the readings rise very fast and can go off the newton-meter scale.



8. Calculate the tension at 20° — what each newton meter should read. Check:  Lab → 500 g on cotton thread, 20° .

Working

Results ($m = 0.500\text{ kg}$, same angle on both sides). Theory: $T = mg / (2 \sin \theta)$.

$\theta / ^\circ$	left reading / N	right reading / N	mean T / N	T from theory / N	$1 / \sin \theta$
60					
45					

30					
20					
10					

9. Cotton thread breaks above 20 N. Calculate the smallest angle at which it can still hold the 500 g mass. Test it in the sim 1° above and 1° below.

Working: $2T \sin \theta = mg$

10. Set 20° on the left and 50° on the right (500 g). Calculate T_L and T_R , then compare with the newton meters and the sim (500 g, *angles 20° and 50°*).

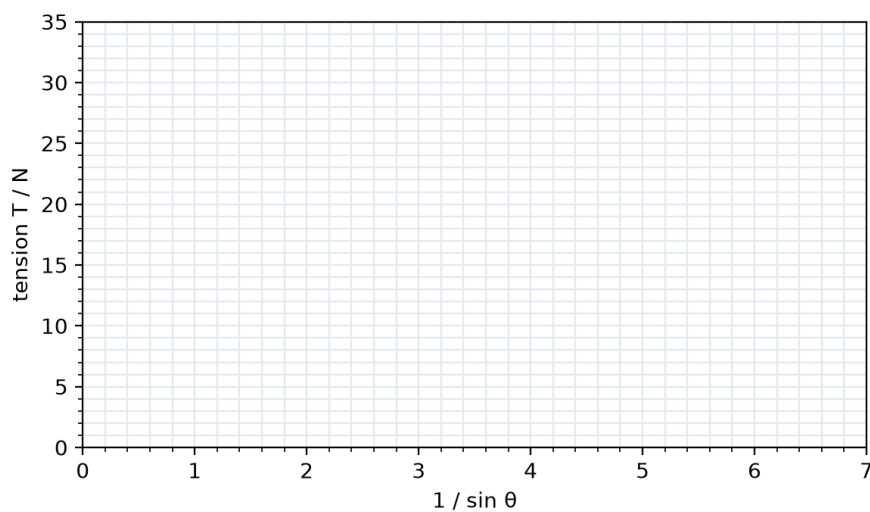
Working: $\Sigma F_x = 0$ and $\Sigma F_y = 0$

11. Explain why the newton-meter readings rise so much when the string is nearly straight.

Explanation

Part 3 — Homework: analyse

12. 🧪 Lab → 1.0 kg, *unbreakable*. Set $\theta = 10^\circ, 20^\circ, 30^\circ, 45^\circ, 60^\circ$ and 80° (same both sides), press ▶ Let go and ● Record each one. Plot T against $1/\sin \theta$ on the grid, draw the best-fit line and find its gradient. Use it to find the mass (gradient = $mg/2$).



Gradient and mass

13. 🧑 A 70 kg tightrope walker makes the wire dip 5.0° on each side. Calculate the tension in the wire and how many times the walker's weight it is.

Working

14. 🧑 A nylon cord breaks above 2.0 kN. Calculate the smallest angle at which it could hold the 70 kg walker.

Working

15. 🚦 A 15 kg traffic light hangs from two cables at 30° (left) and 60° (right) below the horizontal. Calculate the tension in each cable.

Working: FBD $\rightarrow \Sigma F_x = 0$ and $\Sigma F_y = 0$ (or a vector triangle)

16. 🐦 A 30 g bird sits in the middle of a wire that dips 3.0° on each side. Calculate the tension in the wire.

Working

17. 🏮 A 2.0 kg lantern hangs from the middle of a string; each half makes 40° with the horizontal. Calculate the tension.

Working

18. 🏠 (Problem set Q19) A 15 kg sign hangs from two cables, each at 40° to the horizontal. Calculate the tension in each cable.

Working

19. 🧺 3.0 kg of wet washing hangs in the middle of a line pulled tight (4° each side). Fishing line breaks above 100 N. (a) Show that the line snaps. (b) Calculate the smallest angle the line may make with the horizontal.

Working